Surface developments, true shapes & intersections

Higher · Drawing & spatial awareness · Answers

Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Describe two ways the production of the packaging could be made more environmentally friendly.

- Use recycled or sustainable card
- Use card that can be recycled
- Use environmentally friendly (soy or vegetable) inks
- Use less ink
- Reduce the card gauge (thickness)
- Fit as many nets as possible onto each sheet to reduce waste

3 Quick check

- a) A sector of a circle
- b) Reduces waste and cost
- c) An ellipse
- d) π × diameter

4 Key terms

- 1 – C
- 2 – B
- 3 – D
- 4 – A

5 Describe how the curve of intersection is found on the elevation.

- Divide the smaller cylinder into equal strips (generators) in the end view/plan
- Project where each generator meets the surface of the larger cylinder
- Project these points into the elevation to meet the corresponding generators
- Join the points with a smooth curve

6 Explain what is meant by ‘Line of intersection’.

- The line where two joined solids' surfaces meet.

7 Explain what is meant by ‘True shape of an angled cut through a cylinder’.

- An ellipse.